Do Open LLMs Compute or Retrieve? A Multi-Probe Study of Physics Reasoning in Open-Weight Language Models

Anonymous Authors

Abstract

We study whether open-weight language models actually *compute* when they answer multi-step physics problems, or whether they pattern-match against memorised solutions. Rather than reporting a single accuracy number, we combine five probes on SciBench physics (323 problems) with a multiple-choice control (MMLU physics, 488 problems) and a second free-response benchmark (OlympiadBench physics, 351 problems, 152 numerically gradable). Our main findings are: (i) prompting matters more than scale — greedy chain-of-thought roughly doubles accuracy over answer-only for the Qwen2.5 family (2.43× at 7B) while a two-fold parameter increase buys only 4.6 points; (ii) the same 7B model answers 66.6% of multiple-choice physics items but only 21.1% when it must produce the number itself, which rules out missing knowledge as the explanation; (iii) a semantics-preserving paraphrase flips 30.4% of final answers while leaving accuracy unchanged, i.e. the flips are symmetric error jitter, and this holds for every model we tested; (iv) restoring clean residual states recovers behaviour mainly from the early-middle layers (peak at layer 8 of 28), and splitting output execution from semantic correctness shows that parseability is essentially layer-invariant (96% even under corruption) whereas correctness tracks restored depth — recoverable only for problems the model can already solve; (v) prompt-level scaffolding does not substitute for capability (+2.0 points, McNemar $p = 0.700$). An error-type annotation cross-checked by a second, cross-family judge shows that the dominant failure is arithmetic rather than physical, but also that the error taxonomy itself is annotator-dependent ($\kappa = 0.133$), which we report as a measurement limitation rather than hide. All experiments were run on a single 80 GB GPU in about 2.6 GPU-hours; code, per-run metadata and the dual-annotated sample are released.

1 Introduction

Chain-of-thought prompting [10] is routinely credited with giving language models "reasoning" ability, and physics benchmarks such as SciBench [9] and OlympiadBench [3] are routinely used to demonstrate progress. Yet a single accuracy number cannot distinguish two very different internal situations: a model that derives the answer from the stated quantities, and a model that recognises the problem family and emits a plausible-looking number. The distinction matters for anyone who wants to trust, audit or improve such systems.

This paper takes the distinction seriously and treats it as an empirical question with several independent probes, rather than as a claim about model internals. We ask three questions and answer them with measurements that are cheap, deterministic and reproducible on a single GPU:

- **What does performance look like?** (§4.1) Across five open-weight models on 323 SciBench physics problems, with bootstrap confidence intervals and per-sub-domain breakdowns.
- **Is the model computing or retrieving?** (§4.2) We perturb the same problems in three ways that are semantically different but superficially similar: change the numbers, rewrite the wording, and exchange two physical quantities.
- **If it computes, where?** (§4.3) We use corrupt-and-restore causal tracing to localise the computation in depth, and separate "producing a number" from "producing the right number".

Two further probes close the loop. A multiple-choice control (§4.4) holds knowledge fixed and varies only the answer format; an error-type analysis with a cross-family annotator check (§4.5) asks *which* stage

fails and how reliable such attributions are. We also test whether prompt-level scaffolding can repair the instability we find (§4.6).

Our contributions are:

1. A multi-probe protocol that separates computation from retrieval using only black-box generation plus (for one open-weight model) residual-stream interventions, together with the finding that *paraphrase instability* is a reproducible, model-generic signal of surface-conditioned answering.

2. A representation-vs-output decomposition: output execution is layer-invariant while correctness is layer-localised, and failures on problems the model cannot solve are *not* recoverable by restoring any single layer.

3. An honest measurement result: with a same-family and a cross-family judge annotating the same 495 errors, agreement is only $\kappa = 0.133$. We therefore claim only annotator-robust statements and release the dual-annotated sample as a resource for LLM-as-a-judge robustness research.

2 Related Work

Physics and science reasoning benchmarks. SciBench [9] collects college-level textbook problems with free-response answers; we use its physics subsets and restrict to the 323 problems whose reference answers are numerically gradable. OlympiadBench [3] contains competition problems; only 43% of its physics items have numerically gradable references, a coverage limit we carry through all our tables. MMLU [4] provides multiple-choice knowledge questions and serves here as a format control rather than as a target.

Causal tracing and localisation. Activation- and state-level interventions are standard for localising behaviour in open models [5, 8]. We use the simplest variant — corrupt the input, then restore the clean residual stream at a single layer — because it requires no training and yields a per-layer curve. We deliberately restrict this probe to one open-weight model, since hidden-state access is impossible for API-only models, and we report the strict inclusion rule that shrinks the analysable sample.

Sensitivity and robustness of LLM answers. Prompt sensitivity is well documented [7], usually as a nuisance to be averaged out. We instead treat sensitivity as the measurement: a perturbation that leaves physics and data untouched but changes the answer is direct evidence that the answer depends on surface form.

LLM-as-a-judge and error taxonomies. Using a strong model to label errors is common practice [11], and its reliability is usually reported as agreement with human labels. We contribute a case where two *LLM* judges disagree substantially on a subjective taxonomy, and argue that this disagreement is itself the informative quantity.

3 Method

3.1 Benchmarks and grading

All correctness decisions are made by symbolic/numeric comparison. The final answer is extracted from the generation by a precedence-ordered procedure (`ANSWER:` line, taking the last occurrence; then `\boxed{}`; then "answer is X"; then the last number on the last line), normalised for Unicode minus signs, superscripts and thousands separators, and compared to the reference under a 5% relative tolerance (0.01 absolute for near-zero targets).

Reference answers are parsed with a *stricter* parser than model outputs: a reference that is not a single number is marked unscorable and removed from the denominator rather than counted as a model error, because a permissive parser reads expressions such as $x^2 + 2x$ as 2 and silently corrupts the denominator. For SciBench physics all 323 references are numerically gradable, so the denominator is the full set; for OlympiadBench only 152 of 351 are, and we report that coverage everywhere it applies. Grading is covered by 23 self-test cases and 83 unit tests, and an end-to-end oracle check (feed the reference as the model output) yields 100% self-consistency.

Table 1: Accuracy (%) on SciBench physics (323 problems). All reference answers are numerically gradable and all model outputs were parseable, so the denominator is the full 323. Bracketed intervals are bootstrap 95% CIs (10,000 resamples, seed=42) on overall accuracy; sub-domain cells are point estimates. "cot" = zero-shot chain-of-thought, "direct" = answer-only.

Model	Mode	Overall (95% CI)	Sub-domain accuracy (%)					
			Math	Mech	Thermo	Optics	Quantum	General
Qwen2.5-3B	cot	12.1 [8.7, 15.8]	20.0	5.4	12.1	10.6	18.2	9.9
Qwen2.5-3B	direct	5.3 [3.1, 7.7]	2.0	7.1	7.6	6.4	6.1	2.8
Qwen2.5-7B	cot	21.1 [16.7, 25.7]	34.0	21.4	21.2	17.0	15.2	16.9
Qwen2.5-7B	direct	8.7 [5.6, 11.8]	8.0	10.7	6.1	8.5	12.1	8.5
Qwen2.5-14B	cot	25.7 [21.1, 30.7]	40.0	26.8	24.2	23.4	21.2	19.7
Qwen2.5-14B	direct	11.8 [8.4, 15.5]	6.0	17.9	7.6	10.6	30.3	7.0
Llama-3.1-8B	cot	9.9 [6.8, 13.3]	14.0	8.9	12.1	6.4	6.1	9.9
Llama-3.1-8B	direct	7.7 [5.0, 10.8]	6.0	7.1	6.1	8.5	18.2	5.6
R1-Distill-7B	cot	11.1 [7.7, 14.6]	12.0	7.1	10.6	8.5	18.2	12.7

3.2 Models and inference

We evaluate five open-weight instruct models: Qwen2.5-3B/7B/14B-Instruct [6], Llama-3.1-8B-Instruct [1], and DeepSeek-R1-Distill-Qwen-7B [2]. Decoding is greedy with a fixed seed, so every reported number is deterministic. We use the tokenizer's chat template, left padding for batched generation, and the `transformers` backend throughout; vLLM was unavailable in our environment, so no result mixes backends.

3.3 Perturbation probes

Each probe is a deterministic transformation of the problem text:

- **numeric**: scale physical values (physics intact, at most two numbers changed, exponents and power bases protected);
- **wording**: paraphrase the framing only, leaving physics and data bit-identical. This is the only variant whose reference answer remains valid, so it is the only one for which we report a paired accuracy comparison;
- **swap**: exchange two distinct physical quantities, producing a physically inconsistent problem.

Because the correct answer to a *numeric* or *swap* variant is not directly computable, those variants are scored by the distribution of output change, and we report median and p90 rather than the mean (the sensitivity score $|\Delta|/(|base| + \epsilon)$ is dominated by outliers when the base answer is near zero).

3.4 Causal tracing

The numeric quantities of a question are replaced by a semantically empty placeholder; the corrupted prompt is run; then the clean residual stream is restored at exactly one layer at a time, all other layers remaining corrupted. Restoration is applied during prefill only and never to decode steps, and clean and corrupted sequences are aligned from the right because they tokenise to different lengths. Only problems whose clean run was correct *and* whose corrupted run lost score enter the recovery rate, which we report together with the skip counts.

4 Results

4.1 How large is the effect of prompting versus scale?

Table 1 reports SciBench physics accuracy for all five models under both prompting conditions, with bootstrap 95% confidence intervals and per-sub-domain breakdowns.

Prompting dominates scale. For the Qwen2.5 family, chain-of-thought multiplies accuracy by 2.28×, 2.43× and 2.18× at 3B, 7B and 14B, whereas doubling parameters from 7B to 14B adds only 4.6 points

Table 2: Sensitivity of final answers to hard vs. surface perturbations (SciBench physics, 323 base problems, answer-only prompt). *numeric* changes physical values with physics intact; *wording* only rewrites surface phrasing (physics and data identical); *swap* exchanges two distinct physical quantities. "Changed" is the share of answers that differ from the base run. Median / p90 are reported instead of the mean: the sensitivity score is $|\Delta|/(|base| + \epsilon)$ and is dominated by outliers when $|base| \approx 0$.

Model	Variant	Coverage	n parsed	Median sens.	p90 sens.	Changed
Qwen2.5-3B	numeric	305/323 (94.4%)	305	0.500	6.32	77.0%
	wording	240/323 (74.3%)	240	0.000	0.91	33.8%
	swap	36/323 (11.1%)	36	0.840	65.25	83.3%
Qwen2.5-7B	numeric	305/323 (94.4%)	305	0.587	8.96	77.7%
	wording	240/323 (74.3%)	240	0.000	0.95	30.4%
	swap	36/323 (11.1%)	36	0.371	15.06	69.4%
Qwen2.5-14B	numeric	305/323 (94.4%)	305	0.633	7.60	83.9%
	wording	240/323 (74.3%)	240	0.000	0.68	31.7%
	swap	36/323 (11.1%)	36	0.533	6.50	80.6%
Llama-3.1-8B	numeric	305/323 (94.4%)	303	0.082	5.11	57.4%
	wording	240/323 (74.3%)	238	0.000	1.00	24.4%
	swap	36/323 (11.1%)	36	0.335	1.83	63.9%

(from 21.1% to 25.7%). The benefit of chain-of-thought is moreover family-dependent rather than a law of nature: for Llama-3.1-8B the same comparison gives only 1.29× (9.9% vs 7.7%). Family effects can exceed scale effects — the 3B Qwen model (12.1%) outperforms the 8B Llama model (9.9%) — which is why we report a spread of models rather than a single scaling curve. The reasoning-distilled model (DeepSeek-R1-Distill-Qwen-7B, 11.1%) does not beat its same-size instruct sibling, which we attribute to the friction between long reasoning traces and this benchmark's terse numeric answer format.

4.2 Is the model computing or retrieving?

Table 2 reports the three perturbation probes for four models.

Changing the numbers moves the answer most of the time (77.7% of answers change for Qwen2.5-7B), so the input values do participate in the computation: the model is not a lookup table. However, a paraphrase that leaves physics and data bit-identical changes 30.4% of answers with no measurable accuracy effect, and the pattern repeats across models (33.8%, 31.7%, 24.4%). Paired McNemar tests on the paraphrase variant are non-significant for every model ($p = 0.371$, $p = 0.131$, $p = 0.752$, $p = 1.000$), so the flips are symmetric error jitter rather than a difficulty increase. The magnitude of the response to *numeric* changes also tracks accuracy within the Qwen family (77.0% → 77.7% → 83.9%), with Llama-3.1-8B the least responsive (57.4%) and the least accurate. Figure 1 visualises this contrast.

4.3 If it computes, where — and is the output on the same pathway?

Table 3 locates the computation in depth. Of 150 attempted problems, 14 satisfy the strict inclusion rule (134 skipped as clean-wrong, 2 as corruption-flat). Recovery peaks at layer 8 of 28 (+0.84), plateaus over the early-middle layers and collapses after layer 18 (Figure 2).

Restoring a clean state at a layer is only defined for problems whose clean run is correct, so this curve cannot tell us whether layer patching could *rescue* failures. To separate the two questions we instrument all 100 attempted problems and record two outcomes at every restore point: whether the model emits a parseable number (output execution) and whether that number is correct (semantics). Table 4 shows the result, split by whether the clean run was correct.

Parseability is essentially layer-invariant: it is 96% even for the corrupted baseline and stays within 97–100% at every restore point. The model always emits a number; corruption never makes it decline. Correctness, by contrast, depends on the restored depth: for the 10 problems the clean run solves, restoration recovers up to 70% of them, whereas for the 90 problems it cannot, never exceeding 3.3% at any single layer. The two pathways are therefore distinct — and, importantly, the failures are capability limits rather than misplaced representations that a layer patch could repair.

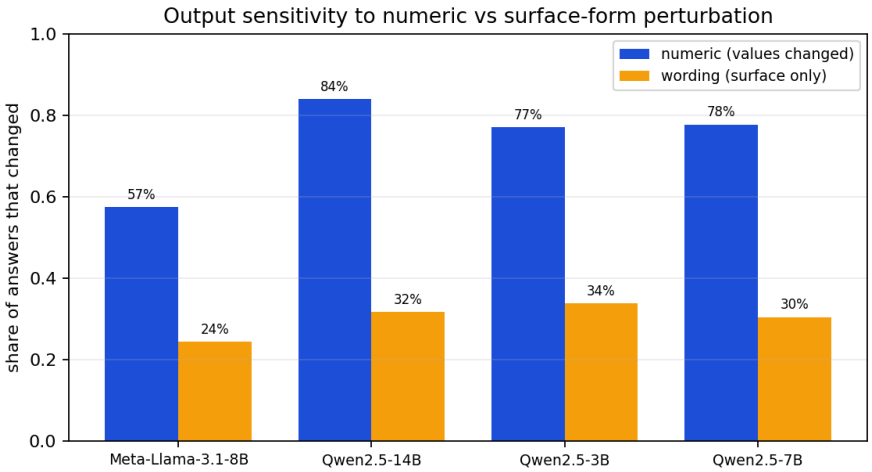


Figure 1: Share of final answers that change when physical values are rescaled (*numeric*) versus when only the phrasing is rewritten (*wording*). The paraphrase bar is non-zero for every model, which is the paper's central robustness observation.

Table 3: Causal tracing on Qwen2.5-7B-Instruct (28 layers): recovery rate when the clean residual stream is restored at a single layer of the corrupted run. Only problems whose clean run was correct *and* whose corrupted run actually lost score are counted ($n = 14$ of 150 attempted; skipped 134 clean-wrong and 2 corrupted-flat).

Layer range	Mean recovery	Mean ±1 std across problems
layers 0–2	+0.431	±0.912
layers 4–8[†]	+0.665	±0.378
layers 10–16	+0.614	±0.431
layers 18–22	+0.238	±0.337
layers 24–26	+0.061	±0.174

[†]Peak recovery occurs at layer 8 ($+0.837 \pm 0.251$). Layers 0–2 show high variance because restoring the embedding-level state is unstable; recovery collapses after layer 18.

4.4 Knowledge or execution? A format control

Table 5 holds the model and the subject domain fixed and varies only the answer format. The same Qwen2.5-7B gets 66.6% of 488 multiple-choice physics items right but only 21.1% when it must produce the number itself. Since the multiple-choice items test the same physics, missing knowledge cannot be the explanation: the knowledge is retrievable, the numerical execution is not.

OlympiadBench provides a second free-response benchmark: accuracy rises from 23.7% to 30.3% with scale (+ 6.6 points), consistent in direction with SciBench, but its 43% gradable coverage is biased towards simpler items and is therefore never compared numerically against SciBench.

4.5 Which stage fails, and can we tell reliably?

We annotated every SciBench chain-of-thought error with a three-way scheme — E1 wrong physical law, E2 algebraic/arithmetic/unit error, E3 violated constraint — and repeated the annotation with a second, cross-family judge on the same samples.

Under both judges E2 dominates and E3 stays small, and the 7B/14B distributions do not differ significantly, i.e. the larger model gets more problems right rather than failing differently. This matches the format control above: the bottleneck is execution, not physical knowledge.

Agreement between the two judges is, however, only $\kappa = 0.133$ (60% raw agreement over 495 samples). The precise E1/E2 split is therefore *annotator dependent*: applying the wrong law usually also produces wrong arithmetic, so the observable text is consistent with either label. We report the coarse, annotator-

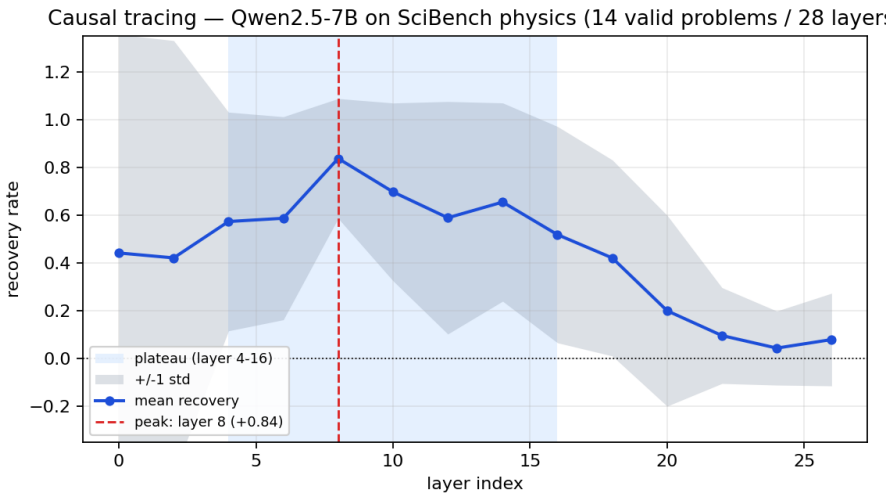


Figure 2: Recovery rate as a function of the restored layer, with ±1 standard deviation across problems. Behaviour is recoverable mainly from the early-middle layers.

robust statement only, and treat the disagreement as a finding about error taxonomies rather than as an inconvenience (§5).

4.6 Can prompt scaffolding repair the instability?

If the paraphrase instability were caused by a missing solution scaffold, telling the model to write down the knowns and unknowns explicitly should help. At a matched token budget on the same 150 problems, it changes accuracy from 24.0% to 26.0% (+2.0 points, McNemar $p = 0.700$) — no reliable gain (Table 7). Prompt-level scaffolding does not substitute for capability.

The same table exposes a separate measurement limitation worth flagging: between 71% and 90% of outputs ignore the explicitly requested `ANSWER:` line and are graded through the fallback extractor. Output lengths stay far below the token budget, so this is a format-compliance failure, not truncation, and accuracy is statistically indistinguishable between compliant and non-compliant outputs.

5 Discussion

What the evidence jointly supports. Four independent probes converge on the same reading of open-weight LLM physics performance. (i) Accuracy on SciBench physics tops out at 25.7% for the largest model we ran and gains only +4.6 percentage points from a two-fold parameter increase, while chain-of-thought prompting roughly doubles accuracy (2.2–2.4× for the Qwen family) — prompting matters more than scale. (ii) The same model reaches 66.6% on multiple-choice MMLU physics but only 21.1% when it must produce the number itself, which rules out the "no physics knowledge" explanation: the knowledge is present and retrievable, the numerical execution is not. (iii) A perturbation that leaves physics and data bit-identical and only rewrites surface phrasing flips 24%–34% of final answers in every model we tested, without changing accuracy on average, i.e. the flips are symmetric error jitter rather than a difficulty effect. (iv) Restoring clean residual states recovers behaviour mainly from the early-middle layers (peak at layer 8 of 28, plateau over layers 4–16) and collapses after layer 18. (v) Splitting output execution from semantic correctness separates the two pathways: parseability is essentially layer-invariant (96% even under corruption, 97–100% at every restored layer), whereas correctness tracks the restored depth — recovering to 70% at layers 0/12 for problems the model can solve, but never above 3.3% at any layer for the 90 problems it cannot solve. Taken together: outputs are produced by computation, but the computation is not a stable physical solver — it is surface-conditioned pattern matching with a fragile arithmetic tail, and the failures are capability limits rather than misplaced representations that a layer patch could repair.

Table 4: Output execution versus semantic correctness. The clean residual stream is restored at one layer of the corrupted run; we record whether the model emits a parseable number (*parse*) and whether that number is correct (*acc*). Parseability is essentially layer-invariant, whereas correctness moves with the restored depth — and only for problems the model can solve without corruption.

Condition	Parse rate	Accuracy	
		clean correct (n=10)	clean wrong (n=90)
corrupted (no restore)	96.0%	–	1.0%
clean (no corruption)	100.0%	–	10.0%
restore layer 0	99.0%	70.0%	1.1%
restore layer 4	99.0%	40.0%	3.3%
restore layer 8	99.0%	50.0%	0.0%
restore layer 12	97.0%	70.0%	0.0%
restore layer 16	100.0%	60.0%	3.3%
restore layer 20	100.0%	30.0%	2.2%
restore layer 24	97.0%	10.0%	0.0%

Qwen2.5-7B-Instruct, 100 attempted problems, 7 restore points (every 4 layers of 28). Parse rate stays within 97.0–100.0% at every layer, including the corrupted baseline: the model always emits a number. Correctness, by contrast, is only recoverable for problems the clean run solves, and never recovers for the rest at any single layer.

Table 5: Same model, different answer formats. Multiple-choice recognition is easy; producing the final number is not. Task formats are not directly comparable (selection vs. generation), so the contrast is interpreted qualitatively.

Task	Answer format	n (gradable)	Accuracy
MMLU physics (high-school + college + conceptual)	select A–D	488	66.6 [62.3, 70.7]
SciBench physics, cot	write the number	323	21.1 [16.7, 25.7]
SciBench physics, direct	write the number	323	8.7 [5.6, 11.8]
OlympiadBench physics, cot	write the number	152 of 351	23.7 [17.1, 30.3]

Model: Qwen2.5-7B-Instruct. OlympiadBench is evaluated only on the subset whose reference answer is numerically gradable (152 of 351), which is biased towards easier items; it must not be compared numerically against SciBench.

Why the error taxonomy is itself uncertain (and why that is a contribution). Our cross-annotator agreement ($\kappa = 0.133$, 60.2% raw agreement over 495 samples) is low, and this is not merely a nuisance for our own error analysis — it reveals a fundamental limitation of the error taxonomy itself: the boundary between "misapplying a physical law" (E1) and "algebraic miscalculation" (E2) is inherently ambiguous, even for strong LLMs. When a model applies the wrong law, the resulting arithmetic is also wrong, so the observable output is consistent with either label; conversely, an arithmetic slip inside a correct derivation is textually indistinguishable from a mis-selected relation. We therefore report only the coarse, annotator-robust statement — E2 dominates and E3 is rare under *both* judges — and deliberately do not quote precise category percentages. This suggests that evaluating AI reasoning purely through coarse-grained error categories is insufficient: taxonomy-based error analysis inherits an annotator-dependent ceiling that is invisible unless agreement is measured. We release our dual-annotated sample (495 wrong answers from two models, each labelled independently by a same-family judge and a cross-family judge) as a benchmark resource not only for physics reasoning, but for future work on the robustness of LLM-as-a-Judge error attribution, where the *disagreement* between judges is the signal rather than noise.

Implications for mitigation. Two of our findings are directly actionable. First, since the dominant failure mode is arithmetic rather than physical (E2), the highest-leverage interventions are execution-side — tool use / code interpreters, or explicit symbolic bookkeeping — rather than more physical pre-training data. Second, since 24%– 34% of answers move under a semantics-preserving paraphrase, single-prompt evaluation understates the variance of a model's competence; reporting paraphrase-consistency alongside accuracy is a cheap and informative robustness metric. We also tested whether the instability is recoverable at inference time by adding structural scaffolding to the prompt (requiring the knowns and unknowns to be written down

Table 6: Error-type distribution of the SciBench cot runs, annotated independently by a *same-family* judge (Qwen2.5-14B) and a *cross-family* judge (Llama-3.1-8B). *unscorable_ref* (reference answer not numerically gradable) and *unparsed_pred* (no parseable answer) samples are excluded. E1 = wrong physical law; E2 = algebraic / arithmetic / unit error; E3 = violated constraint / internal inconsistency; N = grader false negative.

Model	Judge	Errors	E1	E2	E3	N
Qwen2.5-7B	Qwen2.5-14B (same family)	116	29.3%	58.6%	9.5%	2.6%
Qwen2.5-14B	Qwen2.5-14B (same family)	240	22.5%	65.0%	8.3%	4.2%
Qwen2.5-7B	Llama-3.1-8B (cross family)	116	11.2%	80.2%	8.6%	0.0%
Qwen2.5-14B	Llama-3.1-8B (cross family)	240	7.5%	85.4%	7.1%	0.0%

Under the same-family judge the distribution does not differ significantly between 7B and 14B ($\chi^2 = 2.63$, $df = 3$, $p = 0.453$). Cross-family agreement on the same 495 samples is only $\kappa = 0.13$ (60% raw agreement), so the *precise* category shares are annotator-dependent; what is robust across both judges is that E2 dominates and E3 stays small (both judges' modal category: E2).

Table 7: Prompt scaffolding as a substitute for capability. The same 150 problems are evaluated with a matched budget (2048 new tokens) under two prompts: standard chain-of-thought, and a structured prompt that additionally asks the model to write down the knowns and unknowns explicitly. Accuracy does not improve significantly, so the instability documented in Table 2 is not a missing-scaffold problem.

Prompt	Accuracy	95% CI	No `ANSWER:`	Output chars (mean / max)
standard cot	24.0%	[17.3, 31.3]	71%	2377 / 4224
structured (knowns/unknowns)	26.0%	[19.3, 33.3]	90%	2373 / 4639
paired difference: +2.0 pp, McNemar $p = 0.700$				

The last column exposes a separate limitation: most outputs ignore the requested `ANSWER:` format and are graded through the fallback extractor. Output lengths are well below the token budget, so this is a format-compliance failure rather than truncation.

explicitly). At a matched token budget on the same 150 problems it changes accuracy by +2.0 percentage points (McNemar $p = 0.70$): prompt-level scaffolding does *not* substitute for capability, so the mitigation has to be execution-side (tools), not wording-side.

Threats to validity. All numbers come from a single GPU class (A800-80G) and a transformers backend without continuous batching, so wall-clock times are not comparable to vLLM setups, although the accuracy numbers are backend-independent given greedy decoding. Causal tracing is limited to one open-weight model and to the subset of problems that satisfy the strict inclusion rule (14 of 150 attempted), so layer localisation should be read as a robust qualitative pattern (early-middle concentration) rather than a precise per-layer estimate. OlympiadBench results refer to the 43.3% of problems with numerically gradable references and must not be compared numerically with SciBench. Finally, all models are evaluated zero-shot with a single fixed prompt per condition; prompt sensitivity is itself one of the phenomena we measure, so no attempt is made to tune prompts per model.

6 Limitations

All numbers come from a single GPU class and a `transformers` backend without continuous batching, so wall-clock times are not comparable to a vLLM deployment, although greedy decoding makes the accuracies backend-independent. Causal tracing covers one open-weight model and the strict-inclusion subset (14 of 150), so layer localisation is a qualitative early-middle pattern rather than a precise per-layer estimate. OlympiadBench numbers refer to the 43% of problems with numerically gradable references. Error-type shares are annotator-dependent ($\kappa = 0.133$) and are therefore reported coarsely. The paraphrase probe uses deterministic rule-based rewriting; a model-generated paraphrase would be stronger but introduces rewrite quality as a confound. Finally, prompt-level scaffolding was tested with one prompt design at one budget, so its null result is evidence against, not proof of impossibility.

7 Reproducibility

Code and version. All code, evaluation scripts, and analysis utilities are released in the accompanying repository (`https://github.com/380992846-tech/EMPR`) at the exact commit used for this paper (`ec5d0c02c35660a3ac83fc27a1b67de9f0b9df6d`). Defaults live in a single configuration file (`config.yaml`); each analysis stage has a single entry point. Every result file records its own provenance in a `meta` block (`git_commit`, execution environment, `prompt_fingerprint`, `max_new_tokens`, `chat_template`, `wall_time_s`), so any number in the paper can be traced to the run that produced it.

Datasets. **SciBench** physics (`https://github.com/mandyyyyii/scibench`; subjects matter, class, fund, diff, quan, thermo) yields $N_{sci} = 323$ problems, of which mechanics 56, thermodynamics 66, optics 47, math 50, quantum 33, general 71. All 323 reference answers are numerically gradable, so no sample is excluded from the denominator of the main table. **OlympiadBench** physics (`https://huggingface.co/datasets/Hothan/OlympiadBench`; configs `OE_TO_physics_en_COMP` and `OE_TO_physics_zh_CEE`) yields 351 problems, of which only 152 (43.3%) have a numerically gradable reference answer; the remaining 199 are symbolic or multivariate solutions and are excluded from the denominator (reported as `n_unscored`). **MMLU** physics (`cais/mmlu`; high-school, college and conceptual physics) yields 488 multiple-choice problems. All datasets are public and are fetched by `scripts/download_data.py`; SciBench is loaded from the local `data/` copy when present so that the exact same 323 problems are used on every machine (`EMPR_PREFER_HF=1` overrides this).

Models. All models are open-weight and were run from public checkpoints: Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Meta-Llama-3.1-8B-Instruct, and DeepSeek-R1-Distill-Qwen-7B. No closed-weight or API-only model contributes to any reported number. Causal tracing is performed exclusively on an open-weight model whose residual stream is accessible (Qwen2.5-7B-Instruct). Models listed in `config.yaml` but not reported here were not run and no claim is made about them.

Inference settings. Decoding is **greedy** (temperature = 0) with a fixed seed (42), so all outputs are deterministic. Maximum new tokens were 1024 for chain-of-thought prompts (1536 for the reasoning-distilled model, whose chains are longer) and 96 for answer-only prompts (64 for multiple-choice and for the sensitivity probes). Inference used the `transformers` backend on a single GPU in bfloat16 with batches of 16–48; vLLM was not available in the execution environment, so the transformers path was used throughout and no result mixes backends. The tokenizer's chat template is applied to every instruct model (recorded as `meta.chat_template`) and batch generation uses left padding.

Answer extraction and grading. Correctness is judged by symbolic/numeric comparison. The final answer is extracted by a precedence-ordered procedure (`ANSWER:` line, taking the *last* occurrence; then `\boxed{}`; then "answer is X"; then the last number on the last line), normalised for Unicode minus signs, superscripts, `\times` and thousands separators, and compared to the reference under a relative tolerance of 5% (absolute tolerance 0.01 for near-zero targets) via the Python/SymPy evaluator. Reference answers are parsed with a *stricter* parser than model outputs: a reference that is not a single number (up to units) is marked `unscorable_ref` and removed from the denominator rather than counted as a model error, because a permissive parser would read `x^2+2x` as 2 and silently corrupt the denominator. Grading is covered by 23 self-test cases (`python -m src.grading`, 23/23 pass) and by the unit-test suite (83 tests). As an end-to-end check, using the reference answers as model outputs yields 323/323 self-consistent judgements. No LLM judge is used for the primary results.

Statistical protocol. All reported accuracies carry bootstrap 95% confidence intervals from 10,000 resamples with seed = 42. The bootstrap is implemented in pure Python using `random.Random` and order statistics, so the intervals are reproduced *identically* across machines, independent of the NumPy version. Pairwise comparisons use McNemar's test (on paired per-problem outcomes); error-distribution comparisons use an uncorrected Pearson chi-square; agreement between the two error annotators is quantified with Cohen's κ.

Table 8: Compute breakdown for all reported runs (single A800-80G, transformers backend, bf16, greedy decoding). Times are GPU-busy time; model download and orchestration are excluded.

Stage	Runs / scale	GPU time
SciBench, chain-of-thought	5 models × 323	1.05 h
SciBench, answer-only	4 models × 323	0.02 h
OlympiadBench, chain-of-thought	2 models × 351 (152 gradable)	0.58 h
MMLU physics, multiple choice	1 model × 488	0.002 h
Scaffolding comparison	1 model × 150 × 2 prompts (2048 tok)	0.57 h
Sensitivity probes	4 models × 323 × 3 variants	0.10 h
Causal tracing, valid-only	1 model, 150 attempted (14 valid)	0.05 h
Causal tracing, all-attempts	1 model, 100 attempted × 7 layers	0.08 h
Error annotation	2 judges × 2 models × 495 samples	0.17 h
Total	25 runs	**≈2.6 h**

Sensitivity probes. Three deterministic perturbations are applied to the same 323 SciBench problems: *numeric* (scale physical values; physics intact; 305/323 problems affected), *wording* (surface paraphrase only, physics and data bit-identical; 240/323), and *swap* (exchange two distinct physical quantities, creating a physically inconsistent problem; 36/323). Because the correct answer to a *numeric* or *swap* variant is not directly computable, only the distribution of output change is reported for those variants; *wording* keeps the reference answer valid, so it additionally supports a paired accuracy comparison. Median and p90 sensitivity are reported instead of the mean, since the score $|\Delta|/(|base| + \epsilon)$ is dominated by outliers when $|base| \approx 0$.

Causal tracing. We follow the corrupt-and-restore protocol: the numeric quantities of a question are replaced by a semantically empty placeholder (C2 corruption), the corrupted prompt is run, and the clean residual stream is restored at exactly one layer at a time; all other layers remain corrupted, isolating each layer's contribution. Restoration is applied only during prefill (never to single-token decode steps), and the clean and corrupted sequences are aligned from the right because they tokenise to different lengths. Only problems whose clean run was *correct* and whose corrupted run *lost score* enter the recovery rate; of 150 attempted problems, 14 qualify (134 skipped as clean-wrong, 2 as corruption-flat), which is the sample size behind the reported curve.

Hardware and compute. All runs were executed on a single NVIDIA A800-80G PCIe GPU (112 vCPU, 991 GB RAM, 50 GB scratch disk) with bf16 weights. Checkpoints were downloaded one at a time and deleted after use, so peak scratch usage stayed below 30 GB; a single 14B checkpoint needs ≈30 GB free. Total compute for all reported runs is ≈**2.6 GPU-hours**, of which ≈2.2 h is the fourteen evaluation runs (summed from the `wall_time_s` recorded in each result file) and the remainder is sensitivity probes, causal tracing, the scaffolding comparison and error annotation. Table 8 gives the breakdown. An additional ≈1.0 GPU-hour was spent on one full-scale causal-tracing run that was aborted (it was too slow to be practical at full layer resolution) and then re-run at reduced scale; it contributes no reported number.

Reproducibility checklist.

- Deterministic decoding (temperature 0, seed 42): ✓
- Public datasets + scripted download, pinned local copy: ✓
- Open-weight models, public checkpoints: ✓
- Symbolic/numeric grading with 23-case self-test and 83 unit tests: ✓
- Reference-answer strictness separated from output-extraction leniency: ✓
- Bootstrap 95% CI, machine-independent (pure Python): ✓

- Causal tracing on an open-weight model only: ✓
- Code, config and per-run metadata at a pinned commit: ✓
- Error-type annotation cross-checked with a second, cross-family judge: ✓

Limitations affecting reproducibility and interpretation. (i) The `swap` perturbation covers only 36/323 problems (11.1%) because many problems contain quantities of a single type or carry no unit; this is reported rather than artificially inflated, and the swap results are treated as supporting evidence only. (ii) OlympiadBench is evaluated only on the 43.3% of problems whose reference answer is numerically gradable; that subset is biased towards easier items, so OlympiadBench accuracies must not be compared numerically with SciBench accuracies. (iii) Error-type shares are annotator-dependent: a same-family judge (Qwen2.5-14B) and a cross-family judge (Llama-3.1-8B) agree on only $\kappa = 0.133$ (60.2% raw agreement) on the same 495 samples. Both judges identify E2 (algebraic/computational error) as the dominant category and E3 as rare, which is what we claim; the precise E1/E2 split is not claimed, because the E1/E2 boundary is intrinsically fuzzy (applying the wrong law usually also shows up as an arithmetic error). (iv) Unparsed model outputs are counted and reported (`n_unparsed_pred`), never imputed; on SciBench this count is zero for all reported runs. A related and more consequential limitation is *format compliance*: between 71% and 90% of chain-of-thought outputs ignore the explicitly requested `ANSWER:` line and are graded through the fallback extractor ("last number on the last line"). Output lengths stay well below the token budget (mean ≈2,400 characters, max ≈4,600 against a 2048-token cap), so this is not truncation, and accuracy does not differ systematically between compliant and non-compliant outputs (0.21 vs. 0.25 in the matched subset), i.e. the fallback does not obviously bias the headline numbers. It nevertheless means that any evaluation relying on strict output formatting would measure a lower ceiling than ours. (v) Throughput figures refer to the transformers backend; vLLM would be 5–10× faster but is not used here, so wall-clock times are not comparable across backends. (vi) Prompt-level scaffolding does not substitute for capability: adding an explicit "write down the knowns and unknowns" instruction changes matched-budget accuracy by +2.0 percentage points (McNemar $p = 0.70$, same 150 problems, 2048-token budget), so we do not claim that prompt engineering repairs the instability documented by the paraphrase probes.

Acknowledgements

[TODO: funding / compute providers / reviewers]

References

[1] Aaron Grattafiori et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[2] Daya Guo et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

[3] Chaoqun He, Renjie Luo, Yuzhe Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jie Hu, Xu Han, Yujie Huang, Yudi Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. OlympiadBench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal scientific problems. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. arXiv:2402.14008.

[4] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2009.03300.

[5] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2202.05262.

[6] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[7] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design, or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.11324.

[8] Alessandro Stolfo, Yonatan Belinkov, and Mrinmaya Sachan. A mechanistic interpretation of arithmetic reasoning in language models using causal mediation analysis. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. arXiv:2305.15054.

[9] Xiaoxuan Wang, Ziniu Hu, Pan Lu, Yanqiao Zhu, Jieyu Zhang, Satyen Subramaniam, Arjun Loomba, Shichao Zhang, Yizhou Sun, and Wei Wang. SciBench: Evaluating college-level scientific problem-solving abilities of large language models. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2307.10635.

[10] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, et al. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2201.11903.

[11] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023. arXiv:2306.05685.